Mark Gentry

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SUMMARY

- Security specialist and Intel SGX/TDX SW architect for 10+ years. Member of SW team that brought industry's first HW-rooted TEE, SGX, to market.
- Liaison to Intel security incident response for security issues affecting SGX/TDX. Responsible for implementing security mitigations in SGX/TDX SW, including mitigations for Spectre and other transient execution (microarchitectural) security issues.
- Implement processes assuring the integrity of the attestation process for SGX and TDX, Intel's Confidential Computing technologies.
- Extensive SW design experience working on wide range of products. Ability to lead with six years engineering management experience while remaining hands-on in product development. Solid foundation in cryptography.

Skills: C, C++, cryptography, x86, 8051, DSP, JavaScript, python, PowerShell, PCI, AC'97, HD Audio, chipsets.

EXPERIENCE

INTEL CORP.
Raleigh, NC
2000 – Present

2017 – Present Principal Engineer, Cloud SW Architecture/SW Architect

- **Led SGX SW response to Spectre and subsequently discovered microarchitectural/transient execution issues. Led mitigation efforts when mitigations were implemented in SW, for example, for Spectre variant 1 and LVI (load value injection). In all cases, heavily involved in defining and executing Intel's Confidential Computing (CC) TCB Recovery process.**
- **Recognized gaps in validation of Intel updates containing mitigations for SGX/TDX vulnerabilities. Volunteered to address this by writing test plan (100+ pages), working with and mentoring validation engineers and by leading effort to utilize "attestation infrastructure" additions dedicated to validation.**

- Implemented improvements to SGX/TDX attestation process. The improvements considered the needs of customers, validation requirements and the overall integrity of the process.
- Work with customers, including Microsoft Azure CC engineers, to ensure understanding of how Intel addresses security issues affecting SGX/TDX and the corresponding implications for attestation.
- Provided oversight to Intel validation teams responsible for SGX/TDX validation after recognizing need for improvements.
- Led response to SGX.Fail research that explored the implications of stale attestation results and where the researchers were able to compromise the privacy of a popular block chain. Worked closely with the block chain vendor to utilize additional information in the attestation response to limit the breach. Balanced tradeoffs while addressing the stale attestation result root cause.
- Began work on Intel TDX, including the introduction and specification of an attestation library/service, for execution in the TDX confidential VM, that hides attestation details from SW running in the TD.
- Led the effort to develop tools and to modify existing GCC and Microsoft tool chains to be able to add LVI mitigations without changing source code, which was not viable for LVI. Defined different LVI mitigation options, realizing the performance impact of full mitigations.
- Study academic and Intel-authored papers detailing research on security issues affecting SGX/TDX.
- Completed level 1 (course work, paper reading) of Intel crypto training.

2008 - 2017 SW Architect

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- Wrote spec for SGX Platform Services (100+ pages), chipset-provided features complementing native SGX features. Specifically, the chipset provided trusted time and replay-protection through monotonic counters. This included a novel method for adding the attestation of these complementary features to attestation of the rest of the SGX TCB (trusted computing base).
- Wrote spec for SGX attestation (200+ pages) including specification of the SGX "architectural enclaves" (AEs). (Use of SGX involves the development and use of enclaves, essentially user mode libraries that can remain secure even when the OS, VMM, BIOS, etc. are compromised.) To limit SGX HW and microcode complexity, the Intel-provided SGX AEs implement a large portion of the SGX attestation process.
- Took over PoC for SGX attestation from Intel Labs.
- Took SGX SW – attestation and SDK portions – through Intel security review process.
- Part of small team that implemented PoC for Intel MPX (Memory Protection Extensions). The implementation utilized what is now known as xucode.

2005 – 2008 SW Architect

Worked on VoIP and then "Wake-on-Event" projects. Wake-on-event was a generalization of Wake-on-LAN/Wake-on-ring and utilized a SIP stack running on a microcontroller integrated in the Intel chipset (part of Intel vPro/AMT).

- Implemented STUN and DNS clients for SIP ThreadX/Posix-based Wake-on-Event project.
- Led small team to develop Linux driver for a PCI SLIC card for VoIP project.
- Modified HD Audio class driver to work with HD Audio SLIC device.
- Wrote HD Audio spec extension for SLIC devices. Worked with multiple chip vendors on their compliant chips.

2000 – 2005 SW Architect/Engineering Manager

- Led HD Audio soft modem project. Communicated significance of project to organization – laptops to quickly transition to these modems after their introduction. Modem representative on Intel's HD Audio initiative team and on the Intel/Microsoft HD Audio joint development team.
- Identified and helped resolve issues in HD Audio bus driver. Brought up HD Audio modem codec, found workarounds for codec bugs, and helped integrate workarounds in bus and modem drivers.
- Responsible for getting modem included in Intel's Mobile Products Group (MPG) reference motherboard kits.
- Ported HD Audio modem driver to 64-bit architecture (amd64, EM64T). Responsible for making sure organization had latest 64-bit Windows builds and 64-bit Intel platforms.
- Technical lead on embedded V.92 modem project where the goals were 1) to switch to have Intel manufacture the modem silicon and 2) to have this silicon support both controller-based and controller-less modems.
- Port of existing modem controller from a Windows driver implementation to an 8051 implementation. Responsible for development of new platform-dependent code for controller-less mode.
- Responsible for selection of development tools. Responsible for debugging methodology, consisting of debug monitor, use of debug UART, and use of logic analyzer and offline analysis of program traces.
- Worked closely with evaluation board designer to ensure board included necessary features like shadow RAM, ISA interface, and two different AFEs. Responsible for design/programming of Lattice FPGAs on evaluation board.
- Early reviewer of HD Audio spec, paying close attention to design of modem codecs. Helped IHVs with design of their HD Audio modem codecs.
- Designed automated modem test, including simple network protocol to allow sharing of modem test equipment.

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2000 – 2002 SW Architect

- Work with Dallas 80C320 ICE prior to availability of new, Intel-manufactured modem chip (described above).
- Worked with lead chip designer on the design of the new modem chip, initially to define/improve programming interface and later to address testbench failures.
- Worked with design teams to ensure evaluation boards for new modem chip met chip validation and development requirements. Drove evaluation board project to make sure we had boards as soon as chips were available.
- Wrote 8051 code for new modem chip's boot ROM. This code was responsible for detecting bondout (controller-based or controllerless/PCI), detecting and copying contents of EEPROM, and setting PCI Subsystem IDs.
- Defined API designed to hide HW-dependent and OS-dependent details. Wrote corresponding code for both bondouts (controller-based and controllerless/PCI) of the new modem chip.
- Developed bringup Windows drivers for the new modem chip: read/write registers, read/write chip memory, test DMA, timers, and GPIO, and load DSP and controller code.

HUGHES NETWORK SYSTEMS
Gaithersburg, Maryland

1999 – 2000 Principal Engineer

- Responsible for video and audio SW development for Hughes's version of the AOLTV settop box, which combined the standard DirecTV box with HW and SW required to browse the web on a TV.
- Used IGST CyberPro5055, a media processor that integrated video, audio, and graphics. Quickly developed video knowledge needed to help debug multimedia HW subsystem. Required to understand advantages and disadvantages of using synclocked video and how to use the video capture and playback engines on the 5055.
- Responsible for making tradeoffs between video quality and graphics quality. Involved deciding how to combine graphics and video while maintaining acceptable quality of each.
- Responsible for implementing closed-captioning feature. Utilized existing communication scheme to convert closed-captioning text available in satellite signal into NTSC encoder input and eventual decoding by TV.
- Used Tornado (VxWorks, GNU C/C++) development environment and remote debugging.

CIRRUS LOGIC INC. (CRYSTAL SEMICONDUCTOR CORP.)
Austin, Texas
1992 – 1999

Developed PC audio products for Crystal. DSP and microcontroller firmware, Windows audio driver, and other SW development/management for seven generations of Crystal audio parts. Responsible for compatibility of these products with industry standards. Products designed in at several OEMs including Dell, HP, and IBM. Chip bringup and HW debugging using logic analyzers, scopes, etc.

1996 – 1999 Audio SW Manager

- Managed group responsible for implementation of new audio features on Crystal's DSP-based parts. Features include DLS wave-table synthesis, 3D audio, four-channel support, EQ, and audio effects. Group was also responsible for defining DSP resource (memory, MIPs) management policy.
- Developed and managed development of Windows drivers (wave, mixer, MIDI, DirectSound) for PCI audio accelerators.
- Developed DSP code for Crystal's proprietary DSP and helped define HW architecture to make PCI audio parts compatible with DOS (legacy) games.

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- Led effort to have PCI audio parts designed into embedded systems (auto, set-top). Wrote specs, developed sample code, and worked with customers to do this.
- Developed and managed development of Windows drivers (wave, mixer, MIDI, DirectSound) for ISA CODECs.
- Programming classes at University of Texas.

1994 – 1996 Embedded Audio SW Project Lead

- Managed development of all 8052 firmware in CS4232, CS4236, CS423xB, and CS4235/9. All these chips included 8052 cores, stereo audio codec, and ISA interface. Worked closely with chip designers to determine size of patch RAM, deliver ROM code, and add features like mapping codec registers into 8052 SFR space.
- Salvaged production CS4232 parts by working around HW problems in firmware.

1992 – 1994 Multimedia SW Engineer

- Responsible for compatibility with HW interfaces used by DOS games (SoundBlaster and MPU-401) on ISA parts and for Plug and Play and APM (advanced power management) compatibility on ISA parts. ISA parts included 8052 cores that executed code to provide this compatibility. Developed this code and defined associated HW/SW architecture. Extensively reverse-engineered games to implement required, compatible HW interfaces.
- Defined interfaces between 8052 and ISA bus and 8052 and CODEC core on Crystal's ISA CODECs.
- Wrote diagnostics SW in C and x86 assembler for the CS4231 and CS4248. SW tests complete functionality, performance, and feature-sets of these parts and were used extensively by Crystal chip and board designers as well as by Crystal customers.

COMPAQ COMPUTER CORPORATION
Houston, Texas
1991-1992 Systems Engineer (Graphics Development)

- Used FPGAs for internal design projects. Designed EISA analysis board to be used for HW and SW profiling of graphics activity. Partitioned design between MACH devices, standard TTL logic, and specialized logic.
- Evaluated local-bus graphics architectures for 486 bus, VESA Local (VL) Bus, PCI, and Compaq host bus.
- Designed simple audio circuits using Analog Devices A/D and D/A converters and wrote required low-level SW while working on Business Audio project.

RESEARCH PLANNING INC.
Arlington, Virginia
1989 SW Engineer

- Compared implementations of signal processing algorithms on “EMSP” HW with C program or with differing HW implementations. Wrote UNIX scripts to automate this testing.

EDUCATION

1990 GEORGIA INSTITUTE OF TECHNOLOGY
Atlanta, Georgia

- Graduated with Master's Degree in Electrical Engineering in December, 1990. Maintained a GPA of 3.9/4.0 while working as a graduate teaching assistant and while holding two part-time jobs.
- Concentrated on digital signal processing, communication, and computer architecture courses. Completed projects in speech and image processing and adaptive filtering.

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1983-1988 **UNIVERSITY OF MARYLAND**
College Park, Maryland

- Graduated Magna Cum Laude in Electrical Engineering in December 1988 and satisfied cooperative education requirement by working with IBM.
- Achieved an overall GPA of 3.8/4.0 while taking courses on control theory, microprocessors, digital logic, digital signal processing, and analog and digital communication.
- Received Undergraduate Research Participation Award in July 1988 to conduct research on neural networks.